Rongfeng **Cheng** | Medical AI Research

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0009-0002-3021-9647

Research Interests

Trustworthy medical AI, ECG and physiological-signal learning, model auditing and external validation, multimodal representation learning, and clinical NLP.

Education

South China University of Technology (SCUT) **Guangzhou, China**
B.S. in Big Data Management and Application *2023–2027*
GPA: 3.46/4.0.
Selected AI coursework: Large Models and Generative AI (93/100), Large Language Models and Prompt Engineering (93/100), Machine Learning (92/100), and Deep Learning (88/100).

Research Experience

AI Research Institute, Guangdong No. 2 Provincial People's Hospital **Guangzhou, China**
Evidence-Use Auditing for Deep ECG Models *Jan 2026–Present*
Advisors: Haobin Zhang and Weibin Cheng.

- Solely led study design, implementation, experiments, analysis, and manuscript preparation; developed a matched-control perturbation audit spanning clinically defined evidence units, operator comparisons, and frozen-weight external transfer.
- Evaluated five diagnoses and three raw-signal models on CODE-15% and PTB-XL; the audit classified 6 of 15 model-diagnosis pairs as evidence-supported and showed that matched controls or operator choice can overturn target-only conclusions. See Manuscript [1].

AI Research Institute, Guangdong No. 2 Provincial People's Hospital **Guangzhou, China**
DLEF: Dataset-Label Evidence Fingerprints *Jun 2026–Present*
Advisors: Haobin Zhang and Weibin Cheng.

- Solely led study design, implementation, experiments, analysis, and manuscript preparation; proposed directional evidence fingerprints for characterizing dataset-label transfer risk.
- Across CODE-15%, PTB-XL, and Chapman-Shaoxing, DLEF distance was associated with label-level external AUROC degradation (Spearman $r = 0.843$ across 108 points; $r = 0.892$ after source-target-label aggregation). See Manuscript [2].

AI Research Institute, Guangdong No. 2 Provincial People's Hospital **Guangzhou, China**
PathCLIP: Evidence-Aware ECG–Text Representation Learning *Jul 2026–Present*
Advisors: Haobin Zhang and Weibin Cheng.

- Leading experimental implementation for ECG–text alignment and zero-shot recognition, including matched-count, permuted-hierarchy, residual-only, and held-out-label controls that separate knowledge effects from added supervision or capacity.

Manuscripts & Publications

[1]: **R. Cheng et al.** "A Matched-Control Perturbation Audit Framework for Evaluating Evidence Use in Deep Learning Models: An Application to 12-Lead ECG Classification." Under review, *IEEE JBHI*. First author.

[2]: **R. Cheng et al.** "DLEF: Dataset-Label Evidence Fingerprints for Explaining Cross-Dataset Performance in ECG Classification." Under review, *IEEE BIBM 2026*. First author.

[3]: "CodeSafetyBench: Evaluating and Improving Ethical Safety in Large Language Model Code Generation." *iScience*, 2026. Co-author. doi:10.1016/j.isci.2026.115073.

[4]: "Artificial Intelligence Applications in X-ray Coronary Angiography: A Comprehensive Review of Image Analysis Techniques." HNNDL 2026. Co-author. doi:10.1145/3795892.3795917.

[5]: "Artificial Intelligence for Echocardiographic Assessment of Pulmonary Hypertension and Right Heart Function: Advances in Intelligent Diagnosis and Risk Stratification." HNNDL 2026. Co-author. doi:10.1145/3795892.3795918.

Clinical Research & Engineering Experience

AI Research Institute, Guangdong No. 2 Provincial People's Hospital **Guangzhou, China**
Research Intern *Nov 2024–Present*

- Solely developed a standardized-patient Q&A pipeline from approximately 8,000 cardiovascular case PDFs, covering LLM-based information extraction, entity disambiguation, Neo4j knowledge representation, and hybrid retrieval.
- Led full-stack and clinical-workflow development for the five-person Dingbei Meimei acne and facial-health assessment team; supported its initial release when the hospital's Refractory Acne Diagnosis and Treatment Center opened in July 2026.

Additional Research Projects

Project Lead, SCUT Student Research Program **Guangzhou, China**
Myocardial Ischemia Knowledge Graph *2025*
Advisor: Xunde Dong. Led a five-student interdisciplinary team in building a domain knowledge graph with collaborative ontology management, XGBoost-based implicit reasoning, and graph-mapped interpretability analysis; completed the program successfully.

Independent Research
IHD Trajectory Similarity Retrieval
Developed trajectory retrieval and post-cutoff validation for ischemic heart disease ICU stays in MIMIC-IV using time-windowed clinical variables and precomputed k-NN similarity graphs.

Patents & Software IP

Patent: **Method, System, and Electronic Device for Medical Question-Answering Output Assurance** – First Inventor; application #202610582199.6; filed Apr 2026; preliminary examination passed.

Software IP: Knowledge Graph Ontology Intelligent Management System V1.0 (#2025SR2454864); LLM-Based Cardiovascular Proactive Health Management Intelligent System V1.0 (#2025SR2080673).

Honors

- First Prize, 2025 Challenge Cup (Jinan University Campus).
- Second Prize, 2025 9th Guangzhou Mathematical Modeling University League.
- Third Prize, 2025 National Mathematical Modeling Contest – Guangdong Division.

Skills

Modeling: PyTorch, XGBoost, representation learning, multimodal learning

Evaluation: Perturbation analysis, matched controls, cross-dataset evaluation

Biomedical Data: 12-lead ECG, MIMIC-IV / EHR cohorts, clinical NLP, knowledge graphs

Engineering: Python, TypeScript, SQL, Next.js, Flask, Neo4j, Git, Docker

Languages

Chinese (native) Cantonese (native) English: IELTS 6.5, CET-6 528